Fairy Tales Re-imagined

by

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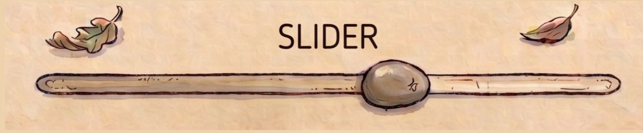
Initial Situation

- Classic Fairy Tales
  - Familiar
  - Experienced in a passive way
    - reading, listening & watching
- Starting Point
  - make Fairy Tales more interactive & playful
- Example: *The Bremen Town Musicians*
  - strong journey structure
  - clear characters
  - natural moment where the animals work together

Analysis of the Usage Context

- Target Group
  - Children (6-10 years)
  - Families
  - Fairy Tale enthusiasts
- Context
  - educational or playful storytelling setting
  - users explore a known story in a new way
- User Goal
  - experience a fairy tale actively instead of only reading it
- Interaction style
  - Buttons to continue story & choose path
  - Light sensor as start trigger
  - Slider for a mini-game
- Visuals show users where they are in the story
- Experience:
  - playful, understandable, and connected to the fairy tale world.



Design Challenge

How can we redesign a classic fairy tale into an interactive experience where users influence the story through physical interaction?

- Challenge
  - combine narrative, visuals, hardware, and interaction into one coherent prototype
- Intent
  - choices to have consequences
  - story stays understandable for children

Main Design Challenge

How can we make users feel that their decisions matter while still guiding them through a clear fairy-tale structure?

Product Concepts and the Selection of the Solution

- Concept Idea 1
  - fairy-tale station
  - physical objects, coins, or figures
  - unlock different stories and paths
- Concept Idea 2
  - branching story interface
  - users make choices at important moments
  - story changes depending on choices
- Development Phase
  - Solar2D for testing interaction
    - Less suitable
- Final solution combines:
  - p5.js visual story interface
  - Arduino-based physical controls
  - story choices with good, neutral, and bad outcomes
  - light sensor for starting key moments
  - slider-controlled mini-game
  - local host server
- Result
  - manageable prototype
  - keeping the interaction physical & playful

Test Results

- What worked well
  - button-based choices
  - story structure with different endings
  - light sensor
  - slider mini-game

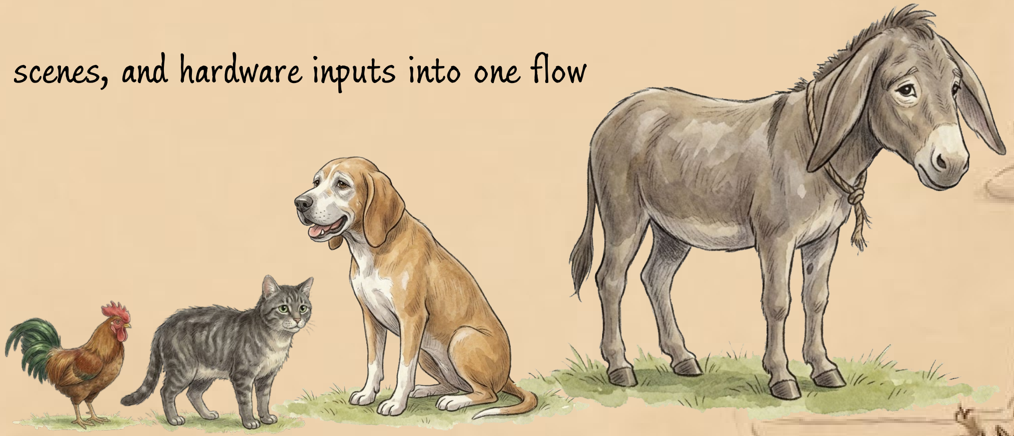
- What needed improvement
  - light sensor threshold calibration
  - hardware button mapping
  - text lengths & legibility
  - mini-game testing
  - physical setup finetuning

Short Demo of the Latest Status

- Current prototype
  - browser-based
  - p5.js application
  - connected to Arduino Micro
- Latest Status
  - visual scenes
  - branching logic
  - keyboard fallback
  - serial input structure
  - mini-game

Reflection

- interaction should support the story
- ideal for multi-use to explore all possible paths of a story
- Testing
  - p5.js
  - Solar2D
  - technical requirements
- continue developing the project in p5.js
  - hosting locally for final setups
- biggest challenge
  - connecting story logic, visual scenes, and hardware inputs into one flow
- next steps
  - polish the p5.js prototype
    - calibrate the hardware
    - improve transitions
    - add audio or voice lines
    - run final user testing



Thank you for your Attention!